

Deep Learning-Based Credit Card Fraud Detection

Project Overview

This project focuses on detecting fraudulent credit card transactions using deep learning techniques. The primary objective is to accurately identify fraudulent transactions while handling severe class imbalance in the dataset.

The project implements:

- Multilayer Perceptron (MLP)
- Autoencoder-based anomaly detection
- Logistic Regression baseline model

The dataset used is the publicly available Credit Card Fraud Detection dataset from Kaggle.

Project Structure

```
credit-card-fraud-detection/  
├── README.md  
├── requirements.txt  
├── environment.yml  
├── .gitignore  
├── credit_card_fraud_detection.ipynb  
├── report.pdf  
├── data/  
│   └── README.md
```

Setup Instructions

1. Clone the Repository

```
git clone https://github.com/gmuttavarapu/credit-card-fraud-detection.git
```

2. Navigate to Project Directory

```
cd credit-card-fraud-detection
```

3. Install Dependencies

Install all required Python libraries using:

```
pip install -r requirements.txt
```

Dependencies

The project uses the following libraries:

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-learn
 - TensorFlow / Keras
 - Jupyter Notebook
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Dataset Instructions

The dataset is too large to upload to GitHub.

Download the dataset from:

<https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>

After downloading:

1. Extract the ZIP file
2. Place `creditcard.csv` inside:

`data/creditcard.csv`

Run Instructions

1. Open Jupyter Notebook

```
jupyter notebook
```

2. Open the Notebook File

Open:

`credit_card_fraud_detection.ipynb`

3. Run All Cells

Run all notebook cells sequentially to:

- Load dataset
 - Preprocess data
 - Train deep learning model
 - Evaluate model performance
 - Generate visualizations
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Model Architecture

The deep learning model consists of:

- Dense(64) + ReLU activation
 - Dropout layer
 - Dense(32) + ReLU activation
 - Dropout layer
 - Dense(16) + ReLU activation
 - Output layer with Sigmoid activation
-

Evaluation Metrics

The following evaluation metrics are used:

- Recall
 - Precision
 - F1-score
 - ROC-AUC Score
 - Confusion Matrix
-

Results Summary

The weighted MLP model achieved strong fraud detection performance:

- Recall: 0.88
- Precision: 0.90
- ROC-AUC: 0.97

The project demonstrates that deep learning techniques significantly improve fraud detection compared to traditional machine learning approaches.

Author

Gopinadh Muttavarapu
COMP 690